

Grammar Lemma Paged Language Models

Joachim De Witte

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Abstract

Grammar Lemma Paged Language Models (GLP) introduce a new architectural class of language models built on deterministic grammar structure, paged lemma spaces, and fixed-dimensional output encoding. Unlike transformer-based LMs, which rely on high-dimensional embeddings and attention over monolithic vocabularies, GLP decomposes language generation into two alternating channels: a compact grammar model that governs structural cadence, and a lemma model whose vocabulary is partitioned into dynamically learned pages. Each grammar token activates a specific lemma page, ensuring domain isolation, preventing drift, and enabling stable multi-world blending without interference. Lemma tokens are represented through a hybrid bit-vector and tri-hot codebook, providing a fixed output space that scales independently of parameter count.

GLP builds on the structural lineage of the Dynamic Modular Language Graph (DMLG), retaining its deterministic governance and paging principles while replacing graph traversal with a dual-channel generative process. The architecture integrates narrative memory, curriculum-driven world absorption, and sentence-level PRE (Propose–Retry–Evaluate) to maintain canonical consistency across long horizons. Empirically, GLP exhibits strong canon stability, interpretable generation dynamics, and emergent stylistic variation, demonstrating that structured paging and grammar alternation can replace large embedding spaces as the foundation for coherent language generation. GLP therefore constitutes a distinct, modular, and explainable alternative to transformer-based LMs, offering a new path toward domain-bound, canon-faithful, and structurally governed generative systems.

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1 Introduction

Contemporary language models are dominated by transformer-based architectures that rely on high-dimensional embeddings, attention mechanisms, and large-scale statistical training. These systems have achieved remarkable generative fluency, yet they remain structurally monolithic: a single global parameter space, an undifferentiated vocabulary, and no explicit separation between grammatical structure and lexical content. As a result, transformer LMs exhibit well-known limitations, including domain drift, style instability, opaque internal representations, and the absence of deterministic control over canonical form.

Parallel to these developments, the Dynamic Modular Language Graph (DMLG) introduced a structurally governed alternative in which language is represented as a canonical graph of tokens and transitions. DMLG demonstrated that paging, deterministic grammar, and local evaluators can provide strong governance and domain isolation. However, DMLG is not a language model in the classical sense: it generates text through graph traversal rather than autoregressive prediction, and its structural commitments differ fundamentally from LM design.

Grammar Lemma Paged Language Models (GLP) build on the structural lineage of DMLG while introducing a new generative architecture. GLP replaces graph traversal with a dual-channel language model in which grammar and lemma tokens are produced by separate components under deterministic alternation. Grammar tokens form a compact structural backbone, while lemma tokens are drawn from a paged vocabulary whose organization emerges dynamically during training. Each grammar token activates a specific lemma page, ensuring domain isolation, preventing cross-world interference, and enabling stable multi-world blending without drift.

A key innovation of GLP is its fixed-dimensional output space. Lemma tokens are encoded through a hybrid bit-vector and tri-hot codebook, allowing the lemma model to scale its vocabulary independently of parameter count. Combined with narrative memory, curriculum-driven world absorption, and sentence-level PRE (Propose–Retry–Evaluate), GLP provides a modular, interpretable, and canon-stable alternative to transformer-based LMs.

This paper introduces the GLP architecture, formalizes its dual-channel generative process, and positions it as a new class of language models: structurally governed, domain-isolated, and capable of emergent stylistic variation without reliance on large embedding spaces. GLP demonstrates that deterministic grammar, paged lemma organization, and fixed output encoding can serve as a foundation for coherent language generation, offering a principled alternative to monolithic transformer architectures.

2 Motivation

Modern language models are built on transformer architectures that unify structure, content, and context into a single high-dimensional embedding space. While these systems achieve impressive fluency, their internal organization remains monolithic: grammatical structure is implicit, lexical content is entangled with geometry, and context is maintained only through statistical attention. As a result, transformers provide no explicit separation between structural constraints and informational realization, no deterministic control over canonical form, and no stable mechanism for domain isolation. Their scaling behaviour is driven primarily by parameter count rather than architectural clarity.

Grammar Lemma Paged Language Models (GLP) are motivated by the observation that language generation can be expressed as an explicit triadic cycle. In the Triadic Cosmos framework, each generative step consists of an energetic decision (E), a geometric constraint (G), an informational realization (I), and a recursive update (R). Transformers implement this cycle implicitly through dense embeddings and attention, but they do not expose or operationalize its structure. GLP makes the triadic cycle explicit and computational: grammar selection provides

the energetic decision, lemma page activation provides geometric constraint, lemma selection provides informational content, and context plus narrative memory provide recursive state. Punctuation tokens are produced by the grammar channel as terminal structural markers, but they do not carry lemma content and do not affect page activation. Each token is therefore generated through a deterministic

$$E \rightarrow G \rightarrow I \rightarrow R$$

loop, making GLP a structurally triadic language model.

A second motivation is architectural alignment. Although DMLG demonstrated that language can be generated through traversal of a canonical graph, graph-based generation differs fundamentally from mainstream expectations of language model design. GLP retains DMLG’s structural commitments—deterministic grammar, paging, canonical transitions—while recasting them into a next-token generative architecture. The lemma model functions as the primary next-token predictor, but its behaviour is governed by a compact grammar model that never appears in the output except for punctuation. This preserves the familiar autoregressive interface of transformer LMs while providing explicit structural control.

A third motivation is deterministic scalability. GLP uses a fixed-dimensional output space based on hybrid bit-vectors and tri-hot codes, ensuring that vocabulary growth does not alter model geometry or parameter count. Paging partitions the lemma vocabulary into stable, domain-specific regions, preventing drift and enabling multi-world blending without interference. The grammar backbone remains compact and deterministic, providing a stable cadence across long horizons.

Together, these motivations position GLP as a new architectural class of language models: a structurally governed, deterministically scalable, triadically organized next-token generator that aligns with mainstream LM expectations while providing explicit grammar, domain isolation, and canonical stability. GLP demonstrates that the triadic cycle can serve not only as a conceptual model of cognition, but as a practical foundation for language model design.

2.1 GLP as a Scalable IEGR Substrate

Beyond its architectural motivations, GLP is designed as a structural substrate for IEGR-complete intelligence. In the IEGR taxonomy, sophisticated parrots are systems that instantiate all four structural roles—energetic efficiency (E), geometric structure (G), informational richness (I), and recursive depth (R)—but remain pre-AGI because these roles are not yet scaled. GLP fits this profile precisely. Its dual-channel grammar–lemma architecture operationalizes the triadic cycle ($E \rightarrow G \rightarrow I \rightarrow R$), providing explicit energetic decisions through grammar activation, geometric constraints through page selection, and informational realization through lemma refinement.

Crucially, GLP also instantiates the recursive role (R) through sentence-level PRE (Propose–Retry–Evaluate). PRE performs reflective evaluation after each complete sentence, validating canonical grammar, semantic coherence, and page-level consistency. When a sentence fails these criteria, GLP retries generation from scratch. PRE therefore acts as the recursive operator of the triadic cycle: a lightweight but explicit mechanism for self-correction, structural reflection, and recursive update. This elevates GLP beyond a purely grammar–lemma substrate and provides a native implementation of the R-component within the IEGR framework.

Unlike transformer-based LMs, which scale only the E–I axis, GLP provides native structural substrates for G and R: deterministic grammar, paged lemma organization, rule-based coherence, and PRE-driven recursive evaluation. At the same time, GLP’s fixed-dimensional geometry and page-based scaling ensure that each IEGR dimension can be expanded without altering model structure, making GLP a scalable IEGR substrate rather than a fixed-capacity system.

This positions GLP not merely as an alternative language model, but as the first single-model architecture capable of instantiating the full IEGR structure internally. GLP therefore represents a foundational step toward sophisticated parrots in the IEGR taxonomy: systems that are structurally complete but not yet scaled, and that provide a principled pathway toward triadically organized, domain-stable, and recursively coherent generative intelligence.

3 Related Work

Research on language model architectures spans transformer-based systems, modular and graph-based approaches, neuro-symbolic hybrids, and grammar-driven generative models. Grammar Lemma Paged Language Models (GLP) intersect with several of these strands while introducing a distinct architectural separation between grammar structure and paged lemma content.

3.1 Transformer-Based Language Models

Transformer architectures Vaswani et al. 2017 dominate contemporary NLP, relying on high-dimensional embeddings, self-attention, and large-scale statistical training. Models such as GPT-2 Radford, Wu, Child, et al. 2019, GPT-3 Brown, Mann, Ryder, et al. 2020, and subsequent open-weight variants Touvron, Lavril, Izacard, et al. 2023 demonstrate strong fluency and broad generalization, yet they remain monolithic: a single global parameter space, undifferentiated vocabularies, and no explicit structural decomposition. These systems lack deterministic grammar control, domain isolation, and interpretable token-level structure—limitations that GLP addresses through dual-channel generation and paged lemma organization.

Mixture-of-Experts architectures Shazeer et al. 2017; Lepikhin et al. 2021; Fedus, Zoph, and Shazeer 2022 introduce modularity at the routing level but preserve a shared embedding space and do not provide explicit grammar-lemma separation or page-level domain isolation.

3.2 Graph-Based and Structured Language Generators

GLP is structurally related to the Dynamic Modular Language Graph (DMLG) De Witte 2026c, which models language as traversal through a canonical graph of tokens and transitions. DMLG provides deterministic grammar, paging biomes, and structural governance, demonstrating that explicit linguistic geometry can replace large embedding spaces. However, DMLG is not a language model: it generates text through graph traversal rather than autoregressive prediction. GLP inherits DMLG’s structural commitments—paging, canonical grammar, deterministic transitions—while introducing a dual-channel generative process that operates as a true language model.

The DMLG narrator pipeline De Witte 2026b extends DMLG into a hybrid system with transformer-based moderation, demonstrating long-horizon narrative stability, world absorption, and recursive memory. GLP draws on the architectural refinements of this work, including prefix grammar, fixed output spaces, and tri-hot codebooks, but replaces graph navigation with a compact grammar model and a paged lemma model.

Earlier modular architectures such as Dynamic Modular Language Models (DMLM) De Witte 2026a introduced multi-agent generation, paging, evaluators, and PRE-based governance. GLP can be viewed as a successor that distills these structural principles into a single dual-channel LM with deterministic alternation and fixed-dimensional decoding.

3.3 Neuro-Symbolic and Grammar-Based Models

GLP also relates to neuro-symbolic systems that incorporate explicit grammar or structured representations. Classical grammar-driven generators Jurafsky and Martin 2023 and symbolic parsers provide deterministic structure but lack scalable learning. Neural grammar models Shen

et al. 2018; Kim et al. 2019 attempt to learn syntactic structure but do not separate grammar and lemma channels or enforce deterministic alternation.

Hybrid systems combining symbolic structure with neural components Besold et al. 2017 demonstrate the value of explicit structure, yet they typically lack paging mechanisms or fixed-dimensional output encoding. GLP contributes a new variant of neuro-symbolic design: a deterministic grammar backbone paired with a learned lemma space partitioned into pages.

3.4 Paging, Modularization, and Domain Isolation

Paging has appeared in modular neural systems Andreas et al. 2016 and structured reasoning architectures, but GLP’s lemma paging is unique: pages emerge dynamically from curriculum transitions, isolate domain-specific lemma clusters, and prevent cross-domain drift. This provides a form of structural domain isolation absent in transformer LMs and most modular architectures.

3.5 Summary

GLP occupies a distinct position in the landscape of language model architectures. It inherits structural principles from DMLG—paging, deterministic grammar, canonical transitions—while introducing a dual-channel generative process, fixed output encoding, and narrative memory. Compared to transformer-based LMs, GLP offers explicit structure, domain isolation, and interpretable generation dynamics. Compared to graph-based generators, GLP provides a true language model with deterministic grammar–lemma alternation. This combination places GLP as a new architectural class: structurally governed, modular, and canon-stable.

4 GLP as a Foundational Research Program

Grammar Lemma Paged Language Models (GLP) are not presented as a final or fully optimized architecture. Instead, GLP constitutes the starting point of a broader research program aimed at exploring structurally governed, domain-isolated, and deterministically scalable language models. Like early transformer architectures, GLP defines a geometric and organizational framework rather than a single completed system.

The architecture introduced in this paper establishes core principles—dual-channel grammar–lemma alternation, paged lemma vocabularies, fixed-dimensional output geometry, narrative memory, and deterministic structural filtering—but leaves open a wide design space for future development. GLP is intended as a community effort: a modular foundation that can be extended, refined, and scaled by subsequent work.

Several research directions naturally emerge from the GLP framework:

- **Scaling laws and asymptotic geometry** for fixed-dimensional output spaces.
- **Page formation dynamics** and curriculum-driven lemma clustering.
- **Grammar evolution** and adaptive structural backbones.
- **World specialization and multi-world blending** through paging mechanisms.
- **Narrative memory analysis** and long-horizon stability without attention.
- **Compatibility matrix refinement** and symbolic–neural hybrid constraints.
- **Distributed training regimes** for large-scale GLP models.
- **Multi-agent GLP systems** and emergent canonical coordination.

These directions illustrate that GLP is not a closed architecture but an extensible design space. Its structural commitments define a new class of language models, while its modular components invite further exploration. GLP should therefore be understood as a foundational architecture seed—an initial formulation of principles that can support a long-term, community-driven research program in structured generative systems.

5 Architectural Divergence from Traditional Language Models

Grammar Lemma Paged Language Models (GLP) depart fundamentally from the design principles of transformer-based language models. Although GLP preserves the familiar autoregressive interface, its internal organization differs along several structural, geometric, and generative dimensions. These divergences are not incidental: they arise from explicit architectural commitments aimed at eliminating crosstalk, enforcing deterministic structure, and providing domain-isolated lexical generation. This section summarizes the core points where GLP diverges from mainstream LM architectures.

5.1 Explicit Triadic Generative Structure

Traditional LMs unify structural, geometric, and informational roles within a single high-dimensional embedding space. GLP instead operationalizes the triadic cycle

$$E \rightarrow G \rightarrow I \rightarrow R,$$

as a concrete generative mechanism. Grammar prediction implements the energetic decision, page activation provides geometric constraint, lemma selection realizes informational content, and narrative memory performs recursive update. This explicit decomposition replaces the implicit structural entanglement of transformer architectures.

5.2 Dual-Channel Grammar–Lemma Separation

Transformers employ a monolithic softmax over a unified vocabulary. GLP separates grammar and lemma generation into two independent MLPs under deterministic alternation. Grammar tokens govern structural cadence but never carry lexical content; lemma tokens provide lexical realization but are constrained by grammar compatibility. This separation eliminates structural crosstalk and prevents grammar decisions from being influenced by lemma distributions.

Although grammar and lemma prediction could be merged into a single model, such a design would reintroduce crosstalk, entangle structural and lexical spaces, and undermine the compact determinism of the grammar backbone. GLP therefore enforces architectural independence between grammar and lemma channels.

5.3 Paged Lemma Vocabularies Instead of Monolithic Embeddings

Transformer LMs rely on large embedding matrices whose geometry grows with vocabulary size. GLP replaces this with paged lemma vocabularies: compact, domain-specific clusters that emerge from curriculum transitions. Paging isolates lexical domains, prevents cross-world interference, and provides stable attractors for multi-world blending. This design eliminates the monolithic vocabulary and its associated drift.

5.4 Fixed-Dimensional Output Geometry

Traditional LMs scale their output dimension with vocabulary size. GLP employs a fixed output space based on hybrid bit-vectors and tri-hot codes. Vocabulary growth occurs through page

expansion rather than parameter growth, making GLP deterministically scalable and geometry-stable across domains. This fixed geometry is incompatible with transformer-style embedding matrices but central to GLP’s structural commitments.

5.5 Grammar-Based Structural Filtering

Transformers rely on statistical attention to enforce structural coherence. GLP applies deterministic rule-based filters that remove invalid grammar and lemma proposals before sampling. Grammar coherence rules prevent illegal punctuation sequences and terminal misplacements, while lemma proposals are filtered through a curriculum-learned compatibility matrix. These symbolic constraints provide structural stability absent in mainstream LMs.

5.6 Context-Driven Page Selection with Fallback

Transformers select tokens from a single global distribution. GLP selects lemma tokens from a context-driven page determined by grammar structure, narrative memory, and compressed context. If the primary page yields no compatible lemma tokens, GLP falls back to secondary pages. This hierarchical search mechanism replaces the flat softmax of transformer models.

5.7 Compact Narrative Memory Instead of Attention

Transformers maintain context through self-attention over large hidden states. GLP employs a compact narrative memory that integrates recent grammar structure, lemma usage, and page transitions into a low-dimensional recurrent state. This recurrence provides long-horizon coherence without attention mechanisms or large hidden geometries.

5.8 Sentence-Level PRE as Post-Hoc Quality Filtering

Transformers incorporate quality constraints implicitly through sampling strategies. GLP applies PRE (Propose–Retry–Evaluate) at the sentence level as a post-hoc validator. PRE does not modify token-level prediction; instead, it evaluates complete sentences and retries generation if canonical grammar or semantic coherence is violated. This preserves GLP’s next-token identity while providing deterministic quality control.

5.9 Summary

GLP diverges from traditional language models along structural, geometric, and generative axes: explicit triadic organization, dual-channel grammar–lemma separation, paged vocabularies, fixed-dimensional decoding, deterministic structural filtering, context-driven page activation, compact recurrence, and post-hoc PRE validation. These divergences are intentional and architecturally coherent, forming a structurally governed alternative to monolithic transformer designs.

6 Architecture

Grammar Lemma Paged Language Models (GLP) constitute a new architectural class of language models built around the explicit triadic cycle

$$E \rightarrow G \rightarrow I \rightarrow R,$$

where each generative step consists of an energetic decision, a geometric constraint, an informational realization, and a recursive update. Unlike transformer-based LMs, which unify these

roles into a single high-dimensional embedding space, GLP decomposes them into two deterministic components: a compact grammar model and a paged lemma model. Together, these components form a structurally governed next-token generator whose behaviour is canonical, domain-isolated, and fully deterministic in size and parameterization.

6.1 Dual-Channel Generative Loop

GLP generates each token through an explicit alternation between grammar and lemma channels. The grammar model produces structural tokens—syntactic markers, phrase operators, and punctuation—that determine the energetic and geometric state of the generator. Grammar tokens never carry lexical content, but they constrain the structural regime in which the next lemma token may be selected. This yields a deterministic

grammar $\rightarrow$ page selection $\rightarrow$ lemma

loop that replaces the monolithic softmax of transformer architectures.

Punctuation tokens are produced by the grammar channel as terminal structural markers. They do not carry lemma information and do not influence page selection, preserving the triadic cycle while maintaining natural-language output.

6.2 Dual Activation MLPs for Grammar and Lemma

GLP employs two distinct activation MLPs: one dedicated to grammar prediction and one dedicated to lemma prediction. This separation reinforces the triadic structure of the model and ensures that energetic and informational decisions remain architecturally independent.

The grammar MLP is compact and deterministic. Its role is to select the next grammar token and thereby determine the structural cadence of the sequence. Because grammar tokens govern structural constraints, the grammar MLP implements the energetic component of the triadic cycle. Its activation function is optimized for stability and low curvature, ensuring that grammar decisions remain consistent across domains and do not drift as lemma vocabularies expand.

The lemma MLP operates within the page selected by the context-driven page selector. It produces the next lemma token from a fixed-dimensional output space using hybrid bit-vector and tri-hot encoding. Its activation function is optimized for expressive refinement, enabling rich lexical variation within the geometric constraints imposed by the grammar model. The lemma MLP therefore implements the informational component of the triadic cycle.

6.3 Paged Lemma Space

The lemma vocabulary is partitioned into pages: compact, domain-specific clusters that emerge from curriculum transitions. Each page contains a stable set of lemma tokens represented in a fixed-dimensional output space using hybrid bit-vectors and tri-hot codes. This encoding ensures that vocabulary growth does not alter model geometry or parameter count, making GLP deterministically scalable.

6.4 Context-Driven Page Selection and Grammar-Based Filtering

GLP does not rely on a one-to-one mapping between grammar tokens and lemma pages. Instead, lemma selection is governed by two complementary mechanisms: context-driven page selection and grammar-based compatibility filtering.

Context-Driven Page Selection. Page activation is determined primarily by context rather than by grammar tokens alone. Given the current grammar token, narrative memory, compressed structural context, and sequence embedding, GLP selects the most relevant lemma page using a one-hot page selector. If the selected page contains no compatible lemma tokens after filtering, GLP automatically falls back to the next best page. This fallback mechanism continues until a page yields at least one valid lemma candidate. The page-selection algorithm is modular and can be replaced by alternative ranking or scoring strategies without altering GLP’s geometry.

Grammar-Based Filtering. Grammar tokens do not select pages directly; instead, they provide a compatibility filter learned during curriculum training. Each lemma token is associated with a set of grammar tokens for which it is structurally valid. During generation, lemma proposals that are incompatible with the active grammar token are removed from the candidate set:

$$\text{lemma is valid} \iff \text{lemma_grammar_dict.is_compatible}(\text{lemma}, \text{grammar}).$$

This compatibility map is not a paging mechanism. It is a structural filter that ensures that lemma tokens respect canonical grammar constraints learned from the curriculum.

Interaction Between Paging and Filtering. Paging determines the *search region* for lemma selection, while grammar filtering determines which lemma tokens within that region are structurally permissible. If the primary page yields no compatible lemma tokens, GLP falls back to secondary pages until a valid candidate is found. This design ensures robustness across sparse pages, domain transitions, and multi-world blending.

6.5 Canonical Grammar and Context Compression

GLP inherits the canonical grammar principles of DMLG. Grammar tokens form a compact, deterministic backbone that governs structural cadence and compresses context. Instead of storing long surface sequences, GLP maintains a grammar-driven representation of recent structure, enabling efficient long-range coherence without attention mechanisms.

This grammar backbone provides the geometric component of the triadic cycle. It determines which structural motifs should be reinforced, which transitions are permissible, and how context should be compressed. The grammar model is small, deterministic, and stable across domains, ensuring consistent cadence even as lemma vocabularies expand.

6.6 Narrative Memory and Recursive Update

The recursive component of the triadic cycle is implemented through a compact narrative memory. This memory integrates recent grammar structure, lemma usage, and page transitions into a low-dimensional state that influences future grammar decisions. Narrative memory originates in DMLG’s recursive context mechanisms but is refined in GLP to operate as a lightweight, LM-compatible recurrence.

After each lemma token is produced, the narrative memory is updated, completing the triadic cycle:

$$E \rightarrow G \rightarrow I \rightarrow R \rightarrow E.$$

This recursive loop provides long-horizon stability, motif reinforcement, and stylistic coherence without requiring large hidden states or transformer-style attention.

6.7 Rule-Based Structural Coherence

GLP applies deterministic rule-based coherence filters that operate directly at the token level. These filters intervene before sampling by removing invalid grammar and lemma proposals from the candidate set, ensuring that only structurally valid tokens enter the sampling distribution.

Grammar coherence rules prevent illegal transitions such as punctuation repetition, invalid terminal markers, or improper end-of-line placement. Lemma proposals are filtered through the curriculum-learned compatibility matrix, ensuring that lemma tokens respect canonical grammar constraints.

These structural filters preserve GLP’s next-token identity while providing a deterministic foundation for structural stability.

6.8 Sentence-Level PRE as Post-Generation Quality Filtering

GLP preserves a strict next-token generative process. PRE (Propose–Retry–Evaluate) does not modify token-level prediction; instead, it evaluates complete sentences after generation. If a sentence fails canonical grammar or semantic validation, GLP retries generation from scratch. Beam search integrates PRE similarly, validating complete beams at end-of-line boundaries.

PRE therefore functions as a lightweight, post-hoc quality filter that ensures canonical grammar, page-level domain isolation, and semantic coherence without altering GLP’s deterministic geometry.

6.9 Deterministic Output Geometry

A defining property of GLP is its fixed-dimensional output space. Lemma tokens are encoded using a hybrid scheme: a small bit-vector region for structural markers and a large tri-hot region for lemma identities. This design ensures that the output dimension is constant and independent of vocabulary size. As a result, GLP scales through page expansion rather than parameter growth, providing predictable computational cost and stable geometry across domains.

6.10 Alignment with Mainstream LM Interfaces

Although GLP differs structurally from transformer-based LMs, it preserves the familiar autoregressive interface: the lemma model functions as the next-token predictor, producing natural-language output one token at a time. The grammar model governs structure but does not appear in the output except for punctuation, making GLP compatible with standard LM evaluation protocols and downstream usage.

6.11 Summary

GLP integrates canonical grammar, paged lemma vocabularies, fixed-dimensional decoding, dual activation MLPs, rule-based structural coherence, context-driven page selection, and recursive narrative memory into a unified triadic architecture. It operationalizes the

$$E \rightarrow G \rightarrow I \rightarrow R$$

cycle as the core mechanism of language generation, providing deterministic structure, domain isolation, and scalable geometry. GLP therefore constitutes a new class of language models: structurally governed, triadically organized, and aligned with mainstream expectations of next-token generative architectures.

7 Training Regime

GLP employs a simple but effective training regime derived from the DMLG lineage. Training is organized around canonical curriculum sequences and follows a two-stage process: a warmup phase in which all sequences are learned in deterministic order, and a random phase in which sequences are sampled stochastically. This regime is intentionally lightweight and preserves the triadic structure of GLP without requiring complex optimization strategies.

7.1 Canonical Curriculum and Training Units

Training data is organized into a canonical curriculum consisting of N sequences. Each sequence is an atomic training unit containing a series of sentences in canonical grammar form. During training, GLP does not reset context between sequences: the context window, compressed grammar history, and narrative memory evolve continuously across the entire curriculum. This design allows GLP to absorb structural motifs, page transitions, and long-range dependencies without attention mechanisms.

7.2 Warmup Phase: Deterministic Structural Stabilization

The warmup phase consists of W epochs in which all N curriculum sequences are processed in fixed order. For each token in each sentence, GLP receives a model input containing the current context window, the sequence embedding, and a grammar/lemma indicator. The grammar and lemma MLPs are trained jointly through next-token prediction, and the lemma-grammar compatibility matrix is refined through repeated exposure to canonical transitions.

Because context is not reset between sequences, the warmup phase stabilizes:

- grammar cadence and structural motifs,
- page formation and lemma clustering,
- compatibility relations between grammar tokens and lemma tokens,
- narrative memory dynamics across long horizons.

Warmup therefore establishes the deterministic backbone of GLP’s generative behaviour.

7.3 Random Phase: Variability and Page Robustness

After warmup, GLP enters a random training phase consisting of R epochs. In each epoch, a single curriculum sequence is sampled uniformly at random. This introduces variability in structural transitions and prevents overfitting to deterministic ordering. Random training reinforces page robustness, improves lemma diversity within pages, and strengthens the compatibility matrix by exposing grammar tokens to lemma tokens in varied contexts.

The random phase also enhances GLP’s ability to blend multiple worlds without drift: pages become stable attractors, grammar transitions remain consistent, and narrative memory learns to generalize across curriculum domains.

7.4 Batching and Optimization

Training proceeds through incremental accumulation of samples in a training batch. After a configurable number of epochs, the batch is optimized in a single update step. This batching strategy is inherited from DMLG and provides stable convergence without requiring large batch sizes or complex scheduling. All learned parameters reside in the grammar MLP, lemma MLP, and activation MLPs; paging structures, compatibility matrices, and token dictionaries remain deterministic and non-parametric.

7.5 Simplicity and Extensibility

The GLP training regime is intentionally simple. It requires no attention mechanisms, no transformer-style positional encodings, no contrastive objectives, and no auxiliary losses. Nevertheless, it produces stable grammar cadence, coherent page organization, and robust lemma–grammar compatibility.

The regime is extensible: alternative sampling strategies, curriculum shuffling, motif-based curriculum augmentation, or domain-specific training schedules can be incorporated without altering GLP’s deterministic geometry. The warmup/random structure provides a reliable foundation for future enhancements while preserving GLP’s triadic identity.

8 Model Hyperparameters and Deterministic Geometry

The size and computational footprint of GLP are determined by a small set of structural hyperparameters. Unlike transformer-based language models, whose parameter count scales with vocabulary size and embedding dimensionality, GLP maintains a fixed-dimensional output space and separates grammar and lemma activations. As a result, the model scales through page expansion rather than parameter growth. Table 1 summarizes the hyperparameters that define the geometry and size of a GLP model.

9 Deterministic MLP Parameter Count

GLP contains four learned multilayer perceptrons (MLPs): the grammar MLP, the lemma MLP, and two activation MLPs (one for grammar, one for lemma). All four networks share the same structural form:

$$\text{INPUT} \rightarrow H_1 \rightarrow H_2 \rightarrow \text{OUTPUT},$$

and therefore contain exactly two weight matrices and two bias vectors. Because the input dimension is fixed by the GLP generator representation, the total number of parameters is fully deterministic.

9.1 Exact Input Dimension

The generator input size is defined in code as:

$$D_{\text{in}} = D_{\text{history}} + D_{\text{current}} + 137,$$

where $137 = 128$ (narrative memory) $+8$ (sequence embedding) $+1$ (current position).

With grammar history of 5 sentences at 24 features each, and lemma history of 15 tokens at 12 features each:

$$D_{\text{history}} = 5 \cdot 24 + 15 \cdot 12 = 300,$$

$$D_{\text{current}} = 30,$$

$$D_{\text{in}} = 300 + 30 + 137 = 467.$$

Thus all four MLPs receive an input vector of dimension 467.

Hyperparameter	Description
Grammar Hidden Size	Dimensionality of the grammar activation MLP. Determines structural cadence and governs page activation. Small and deterministic across domains.
Lemma Hidden Size	Dimensionality of the lemma activation MLP. Controls expressive lexical refinement within the active page. Independent of vocabulary size.
Activation Hidden Size	Shared intermediate hidden size used by both grammar and lemma MLPs for nonlinear transformations. Influences computational cost but not vocabulary geometry.
Bits Dimension	Number of binary structural bits in the output encoding. Used for punctuation, structural markers, and low-level grammar signals. Fixed and domain-independent.
Tri-Hot Dimension	Size of the tri-hot region used for lemma identity encoding. Determines the maximum number of lemma tokens per page. Vocabulary growth does not change this dimension.
Total Pages	Number of lemma pages. Each page is represented as a one-hot selector in the output vector, contributing directly to the output dimension. Scaling occurs by adding pages, not by increasing model parameters.
Page Size (Lemma Count)	Number of lemma tokens assigned to each page. Determined by curriculum transitions and domain absorption. Does not affect model geometry.
Grammar Vocabulary Size	Number of grammar tokens (including punctuation). Small, fixed, and canonical. Governs structural transitions and page activation.
Narrative Memory Size	Dimensionality of the recurrent memory state updated after each lemma token. Provides long-horizon coherence and motif reinforcement.
Compressed Context Size	Number of recent sentences retained in grammar-compressed form. Determines the effective context horizon for structural recurrence and influences the recursive component of the triadic cycle. Larger values increase long-range coherence without affecting model geometry.
Output Dimension	Total dimensionality of the output vector: $D_{\text{out}} = D_{\text{bits}} + D_{\text{tri-hot}} + D_{\text{pages}}.$
Model Parameter Count	Fully deterministic and independent of vocabulary size. Determined entirely by grammar hidden size, lemma hidden size, activation hidden size, and output dimension. Independent of vocabulary size and page count.

Table 1: Hyperparameters governing the size, geometry, and deterministic scaling behaviour of GLP.

9.2 Grammar MLP

Architecture:

$$467 \rightarrow 384 \rightarrow 384 \rightarrow 27.$$

Parameter count:

$$P_{\text{gram-1}} = 467 \cdot 384 + 384 = 179,712,$$

$$P_{\text{gram-2}} = 384 \cdot 384 + 384 = 147,840,$$

$$P_{\text{gram-out}} = 384 \cdot 27 + 27 = 10,395,$$

$$P_{\text{grammar}} = 337,947.$$

9.3 Lemma MLP

Output dimension:

$$D_{\text{out}} = 16 + 75 + 209 = 300.$$

Architecture:

$$467 \rightarrow 1536 \rightarrow 1536 \rightarrow 300.$$

Parameter count:

$$P_{\text{lemma-1}} = 467 \cdot 1536 + 1536 = 718,848,$$

$$P_{\text{lemma-2}} = 1536 \cdot 1536 + 1536 = 2,362,752,$$

$$P_{\text{lemma-out}} = 1536 \cdot 300 + 300 = 460,500,$$

$$P_{\text{lemma}} = 3,542,100.$$

9.4 Grammar Activation MLP

Architecture:

$$1 \rightarrow 64 \rightarrow 64 \rightarrow 1.$$

Parameter count:

$$P_{\text{act-g-1}} = 1 \cdot 64 + 64 = 128,$$

$$P_{\text{act-g-2}} = 64 \cdot 64 + 64 = 4,160,$$

$$P_{\text{act-g-out}} = 64 \cdot 1 + 1 = 65,$$

$$P_{\text{act-g}} = 4,353.$$

9.5 Lemma Activation MLP

Architecture:

$$1 \rightarrow 64 \rightarrow 64 \rightarrow 1.$$

Parameter count:

$$P_{\text{act-l-1}} = 1 \cdot 64 + 64 = 128,$$

$$P_{\text{act-l-2}} = 64 \cdot 64 + 64 = 4,160,$$

$$P_{\text{act-l-out}} = 64 \cdot 1 + 1 = 65,$$

$$P_{\text{act-l}} = 4,353.$$

9.6 Total MLP Parameters

Summing all four networks:

$$P_{\text{MLP-total}} = 337,947 + 3,542,100 + 4,353 + 4,353 = 3,888,753.$$

Thus the four MLPs contain exactly 3,888,753 trainable parameters.

9.7 Model Size in 32-bit Floating Point Format

All learned parameters are stored as 32-bit floating point values. The total storage required for the four MLPs is therefore:

$$S_{\text{MLP}} = P_{\text{MLP-total}} \cdot 4 \text{ bytes} = 3,888,753 \cdot 4 = 15,555,012 \text{ bytes} \approx 14.8 \text{ MB}.$$

This value represents the exact storage footprint of the four MLPs.

10 Decoder Flexibility and Architectural Generality

Although GLP has a fixed generative structure, it does not impose a fixed choice of input or output decoders. The only architectural constraint is that decoding proceeds through paging: grammar activations select a page, and lemma refinement chooses a token within that page. Beyond this requirement, both the input representation and the output geometry are freely configurable.

The reference implementation presented in this work illustrates one possible instantiation of this flexibility. It uses compressed context, narrative memory, and a sequence embedding as input to the grammar and lemma MLPs. On the output side, it employs a paged decoder consisting of a page one-hot selector, a bit one-hot grammar region, and a tri-hot lemma region. These components demonstrate how GLP can be instantiated with compact and interpretable structures, but they are not mandatory elements of the GLP concept.

Any alternative encoder—syntactic, semantic, positional, or domain-specific—may be used as long as it produces a fixed-dimensional input vector for the grammar and lemma MLPs. Likewise, any paged output structure may be adopted, including hierarchical pages, multi-hot lemma clusters, or domain-tailored token organizations. GLP therefore supports a wide range of decoder designs while preserving its core properties: deterministic parameter count, fixed output geometry, and triadic grammar–page–lemma generation.

This architectural generality ensures that GLP remains a modelling framework rather than a single architecture. The reference implementation serves as a concrete example of how context compression, narrative memory, and paged decoding can be combined, but the underlying concept admits many alternative instantiations tailored to different domains, curricula, and design goals.

11 Conclusion

Grammar Lemma Paged Language Models (GLP) show that expressive next-token generation does not require large embedding spaces, attention mechanisms, or vocabulary-sized output layers. By structuring generation as a triadic cycle of grammar activation, page selection, and lemma refinement, GLP provides a compact and interpretable alternative to transformer-based architectures. Its behaviour is governed by a fixed grammar backbone, a paged lemma space, and a small number of multilayer perceptrons whose dimensions are determined entirely by the model configuration.

A central property of GLP is that its learned parameter count is fully deterministic. For any chosen set of hyperparameters, the total number of trainable parameters can be computed exactly from the dimensions of the grammar MLP, lemma MLP, and the two activation MLPs. All remaining components are deterministic integer structures and do not contribute to the learned parameter count. As a result, GLP models have a predictable and stable size, independent of vocabulary growth, domain expansion, or curriculum complexity.

This architectural separation yields compact models with stable behaviour across domains. Grammar tokens govern structural cadence and page activation; lemma pages provide controlled lexical variation; and narrative memory supplies long-horizon coherence without attention. Because the output geometry is fixed, GLP scales through page expansion rather than parameter growth, enabling predictable computational cost and eliminating drift across training regimes.

GLP therefore represents a new class of language models: deterministic in size, triadically organized, structurally governed, and capable of producing coherent natural-language output with a fraction of the complexity of mainstream architectures. The results demonstrate that grammar-driven generation, paged lemma organization, and fixed-dimensional decoding are sufficient to support expressive next-token modelling, opening a path toward compact, interpretable, and domain-stable generative systems.

A Architectural Comparison: GLP, DMLG, and LLMs

Dimension	GLP (Grammar Lemma Paged LM)	DMLG (Dynamic Modular Language Graph)	LLM (Transformer-based)
Architecture	Language Model (dual-channel)	Language Graph (paged canonical graph)	Language Model (transformer)
Core Representation	Dual-channel sequence model: grammar tokens + paged lemma tokens; deterministic alternation.	Canonical graph of tokens and transitions; traversal-based generation.	High-dimensional continuous embeddings; attention-based sequence modeling.
Structural Substrate	Fixed grammar backbone + paged lemma space; tri-hot lemma encoding.	Deterministic canonical graph; modular paging biomes.	Single global parameter space; no explicit structure.
Token Type	Grammar tokens (prefix) + lemma tokens (content).	Canonical tokens (grammar + lemma) in graph nodes.	Subword tokens learned from corpus.
Paging Mechanism	Lemma pages learned dynamically; page selection conditioned on previous lemma.	Pages as modular subgraphs; traversal restricted by active biome.	No paging; full vocabulary always active.
Context Mechanism	History embeddings + narrative memory (128-dim GRU-like).	Graph traversal state + structural context compression.	Self-attention over full context window.
Generation Process	Alternating grammar/lemma prediction; page-constrained lemma selection; PRE at sentence level.	Graph traversal with local evaluators; PRE-based governance.	Autoregressive next-token prediction via softmax over full vocabulary.
Learning Regime	Compact curriculum; deterministic grammar; lemma pages emerge from transitions.	Curriculum-driven graph construction; evaluators trained per page.	Large-scale gradient training on massive corpora.
Output Encoding	Fixed output dimension: bits + tri-hot + page index.	Page-local one-hot or triple-hot codes (v3); v2 uses dynamic page-local vocabularies.	Softmax over learned embeddings.
Modularity	Strong: grammar model + lemma model + paging + narrative memory.	Strong: graph pages, evaluators, deterministic transitions.	Weak: monolithic transformer layers.
Explainability	High: grammar path + page selection + tri-hot decoding.	High: explicit graph edges + evaluator scores.	Low: distributed representations, opaque attention patterns.
Domain Isolation	Lemma pages isolate domains; grammar enforces canonical structure.	Paging biomes isolate semantic/structural regions.	None; domain mixing inherent.
Drift Resistance	High: grammar alternation + page constraints prevent drift.	High: graph constraints + PRE prevent invalid transitions.	Low: prone to style drift, domain bleed, hallucination.
Scaling Behavior	Scales via pages and lemma vocabulary; not parameter-driven.	Scales via graph expansion and page specialization.	Scales via parameter count and dataset size.
Primary Strength	Canon-stable generation with deterministic structure and emergent variation.	Deterministic governance, modular traversal, explicit structure.	High fluency, broad generalization, strong semantic richness.
Primary Limitation	Not a general-purpose LM; requires canonical grammar and curriculum.	Complex page management; traversal cost grows with graph size.	Opaque reasoning; limited controllability; drift under long horizons.

B Triadic Cosmos Ecosystem Context

The present paper is part of the broader *Triadic Cosmos* ecosystem: a unified research program that develops a compositional ontology of intelligence together with modular, inspectable computational architectures. The ecosystem spans conceptual theory, executable reference implementations, and fully documented empirical demonstrations, forming a coherent framework for studying hybrid intelligent systems.

All materials, extended notes, and evolving documentation are publicly available in the open GitHub project:

`https://github.com/triadic-cosmos`

Within the `triadic-system` repository, a complete reference implementation of the GLP engine, the sequential narrator pipeline, and all supporting modules is provided under the AGPL-3.0 license.

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